To use our python-based framework, you need to install the GMTSAR software. A full explanation of the installation steps for different computer configurations can be found here:

👉 https://github.com/gmtsar/gmtsar/wiki

For Ubuntu 20.04 LTS or later (I personally recommend using **Ubuntu 22.04** — there is a segmentation fault (core dumped) error in Ubuntu 24.04, which is a Linux bug affecting many software, so if you are using this version, please be very careful), you can follow the guidelines below to install the software.

It should be noted that the following instructions have been successfully tested on several computers without any issues. However, if you encounter any problems during installation, please open a new issue on GitHub. I will try to assist you as quickly as possible.

GMTSAR installation

Run each of the following commands (one by one) in your Ubuntu terminal:

```
sudo apt-get install csh subversion autoconf libtiff5-dev libhdf5-dev wget
sudo apt-get install liblapack-dev
sudo apt-get install gfortran
sudo apt-get install g++
sudo apt-get install libgmt-dev
sudo apt-get install gmt-dcw gmt-gshhg
sudo apt-get install gmt
```

Be sure that you also have **git** and **make** installed:

```
sudo apt install git
sudo apt install make
```

Then

```
sudo -i
cd /usr/local
git clone --branch 6.6 https://github.com/gmtsar/gmtsar GMTSAR
exit
cd /usr/local
sudo chown -R $USER GMTSAR
cd GMTSAR
autoconf
autoupdate
./configure --with-orbits-dir=/tmp CFLAGS='-z muldefs' LDFLAGS='-z muldefs'
make
make install
```

Next, edit your .bashrc file in your home directory and add these lines to the end of it:

```
export GMTSAR=/usr/local/GMTSAR
export PATH=$GMTSAR/bin:"$PATH"
```

Then run:

```
source ~/.bashrc
```

Alternatively, you can add the same lines to your .profile file:

```
nano ~/.profile
export GMTSAR=/usr/local/GMTSAR
export PATH=$GMTSAR/bin:"$PATH"
source ~/.profile
```

For a final assessment of the installation, run the following command:

```
phasediff
```

You should see:

```
phasediff [GMTSAR] - Compute phase difference of two images

Usage: phasediff ref.PRM rep.PRM [-topo topo_ra.grd] [-model modelphase.grd]
(topo_ra and model in GMT grd format)
```

This indicates that GMTSAR has been successfully installed.

Now it’s time to install DefoEye.

1- Other tips

3.1- Help Installing Miniconda and VS Code

First, download the VS Code installer from:

👉 https://code.visualstudio.com/Download

Then install it by right-clicking on the .deb file and selecting Install.

Next, download the Miniconda installer and run the following commands:

```
bash Miniconda.sh
source ~/.bashrc
```

3.2- GMTSAR Troubleshooting (Missing baseline.ps File)

If you encounter a problem in GMTSAR where the baseline.ps file is not displayed, check that you have both awk and gawk installed. For example, I had a similar issue described here:

- https://github.com/gmtsar/gmtsar/issues/887
- https://github.com/gmtsar/gmtsar/issues/1083

To fix it, run:

```
sudo apt-get install gawk
```

(You may see a warning saying that awk is broken — this is not a problem; just ignore it.)

Then set **gawk** as the default:

```
sudo update-alternatives --install /usr/bin/awk awk /usr/bin/gawk 20
sudo update-alternatives --config awk   # choose /usr/bin/gawk
```

Verify the installation:

```
awk --version
```

should say "GNU Awk"

3.3- GitHub Authentication Procedure

```
sudo apt update && sudo apt install gh -y
gh auth login
# Choose: GitHub.com → HTTPS and complete the login process
```

3.4- Important Note About Merging in Parallel

Be careful in the merge parallel section — it is better not to have empty merge_list files, as I have encountered problems related to this.